

Generative AI: Reconstructed Exam

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Partial reconstruction from memory after the exam. Multiple-choice options for objective questions were reconstructed/inferred where the originals were not recalled exactly. Correct answers are in the key at the end.

Negative marking. A wrong True/False answer cancels one correct True/False answer (net -1 relative to leaving it blank). A wrong multiple-choice answer removes $1/3$ of a correct multiple-choice answer. Unanswered questions score zero. Guessing is therefore neutral in expectation; only answer when you have a reasoned basis.

1 True / False

- TF1.** The VAE objective includes a regularization term that encourages a more structured (organized) latent space.
- TF2.** Mode collapse occurs when the discriminator becomes too weak to provide useful gradients to the generator.
- TF3.** ControlNet does not modify the weights of the original (base) diffusion model.
- TF4.** SANA employs an attention mechanism with linear, $O(N)$, computational complexity.
- TF5.** Diffusion models are trained to directly denoise x_t into x_0 .
- TF6.** KID is an unbiased estimator that performs well with small sample sizes, unlike FID.
- TF7.** Classifier-free guidance improves alignment with the prompt, but at excessive intensity it may reduce diversity and introduce artifacts.
- TF8.** In an image-to-prompt problem, two semantically similar textual descriptions may generate very different images, even with the same generator and the same seed.
- TF9.** Linear interpolation between two latent points guarantees semantically valid results.
- TF10.** Latent diffusion reduces computational cost by running the diffusion process in a compressed space instead of operating on pixels.

2 Multiple Choice

- MC1.** What is the key characteristic of DCGAN?
 - (a) Max-pooling for dimensionality reduction.
 - (b) Strided (and fractionally-strided) convolutions instead of pooling.
 - (c) Fully-connected generator and discriminator.

- (d) Self-attention layers at every resolution.
- MC2.** Which metrics are most appropriate for the TP2 image-to-prompt inversion project?
- (a) BLEU and ROUGE between the recovered and reference prompts.
 - (b) FID and KID computed over the dataset.
 - (c) CLIP image-to-image similarity, LPIPS, and pixel MSE.
 - (d) Classification accuracy on the rendered image.
- MC3.** What distinguishes LoRA from full fine-tuning?
- (a) LoRA updates all model weights, but with a smaller learning rate.
 - (b) LoRA freezes the original weights and trains only small low-rank update matrices, whereas full fine-tuning updates all parameters.
 - (c) LoRA requires more memory and compute than full fine-tuning.
 - (d) LoRA quantizes the model to 4-bit, while full fine-tuning keeps full precision.
- MC4.** In generative-model evaluation, what are precision and recall?
- (a) Precision measures coverage of the data; recall measures sample fidelity.
 - (b) Precision measures sample fidelity (samples lying on the real data manifold); recall measures coverage/diversity of the real distribution.
 - (c) Both measure pixel-level reconstruction error.
 - (d) Standard classification precision/recall on predicted labels.
- MC5.** Which classical generative-AI algorithm rewrites rules in parallel to simulate biological growth?
- (a) Iterated Function Systems (IFS).
 - (b) Genetic algorithms.
 - (c) L-systems (Lindenmayer systems).
 - (d) Markov chains.
- MC6.** What is the difference between DreamBooth and Textual Inversion?
- (a) DreamBooth learns a new text embedding while keeping the model frozen; Textual Inversion fine-tunes the model weights.
 - (b) DreamBooth fine-tunes the model weights (with a class/prior-preservation loss); Textual Inversion freezes the model and learns only a new embedding vector for the concept.
 - (c) Both fine-tune the full model and differ only in the number of training images.
 - (d) DreamBooth trains a ControlNet branch; Textual Inversion trains a LoRA adapter.
- MC7.** In a text-to-image model, a reason to use T5 (or another language encoder) in addition to CLIP is:
- (a) T5 is smaller and faster to run than the CLIP text encoder.
 - (b) T5 provides richer language understanding (complex prompts, compositionality) that CLIP's contrastive text encoder lacks.

- (c) T5 encodes images more accurately than CLIP.
- (d) T5 removes the need for a VAE in latent diffusion.

MC8. Which of the following statements about FID is correct?

- (a) FID compares the mean and covariance of deep (Inception) features between real and generated images, assuming Gaussian distributions.
- (b) FID compares real and generated images pixel by pixel.
- (c) FID is an unbiased estimator that works well at low sample counts.
- (d) FID measures only diversity, not fidelity.

MC9. In text generation with LLMs, what does reducing the temperature parameter do?

- (a) Makes the output more random and diverse.
- (b) Sharpens the distribution, making the output more deterministic and focused on high-probability tokens.
- (c) Increases the size of the vocabulary considered at each step.
- (d) Affects only training, not sampling.

MC10. What is GAN inversion?

- (a) Training a GAN in reverse, mapping images back to noise during training.
- (b) Finding the latent code that makes a pretrained generator reconstruct a given real image (mapping a real image into the latent space).
- (c) Inverting the discriminator so it generates instead of classifies.
- (d) Reversing the order of the generator's layers.

3 Direct Theoretical Questions

D1. What is the difference between evaluating image reconstruction, sample quality, and diversity? Name one metric you can use to evaluate each of these.

D2. Why do diffusion models have more stable training than GANs? And why are diffusion models slower at generating samples?

D3. What are the advantages of using MMDDiT instead of a U-Net in models like FLUX and SD3?

4 Long Theoretical Questions

L1. Explain the different ways to control generative outputs, covering the following paradigms:

- Control during training (cGAN).
- Post-hoc analysis of the most significant directions in the latent space.
- GAN inversion and editing methods.
- Personalization methods like DreamBooth and Textual Inversion.

L2. Why isn't image captioning equivalent to prompt inversion? Explain some strategies for prompt inversion, and describe some metrics used to evaluate the candidate prompts in these methods.

Answer Key (objective questions)

True / False

TF1. True. The KL term regularizes each posterior toward the prior $\mathcal{N}(0, I)$, smoothing and organizing the latent space (the very reason prior sampling works at all).

TF2. False. Vanishing gradients arise from a *too-strong* / *saturated* discriminator, not a weak one. Mode collapse is generator-side: G concentrates on a few modes that fool the current D , since nothing rewards coverage.

TF3. True. ControlNet freezes the base model and trains a cloned encoder connected through zero-convolutions; the original weights are unchanged.

TF4. True. SANA uses linear attention (Linear DiT), giving $O(N)$ complexity for efficient high-resolution synthesis.

TF5. False. The standard objective predicts the added noise ϵ (equivalently the score), and denoising proceeds *iteratively* over many steps, not a direct one-shot $x_t \rightarrow x_0$ mapping. (An x_0 -prediction parameterization exists, but the canonical training target is ϵ , applied step by step.)

TF6. True. KID uses an unbiased estimator of the squared MMD, so it is reliable with few samples, whereas FID is biased (especially at low sample counts).

TF7. True. High classifier-free guidance sharpens prompt adherence but reduces diversity and oversaturates / introduces artifacts.

TF8. True. Text-semantic similarity does not imply image similarity: small prompt changes can shift the conditioning and produce very different images, even at a fixed seed.

TF9. False. Linear interpolation in a Gaussian latent passes through low-density regions (the norm collapses), so it does not guarantee valid results; spherical interpolation (slerp) is preferred.

TF10. True. Latent diffusion (LDM) runs the diffusion process in a VAE-compressed latent space rather than in pixel space, cutting compute.

Multiple Choice

MC1. (b) Strided / fractionally-strided convolutions instead of pooling (plus batchnorm, no fully-connected layers, ReLU/LeakyReLU).

MC2. (c) Image-side metrics: CLIP image-to-image similarity, LPIPS, and pixel MSE. A prompt is judged by the image it re-renders, not by text similarity.

MC3. (b) LoRA freezes the original weights and trains only small low-rank update matrices (BA , $B = 0$ at init), so it tunes a tiny fraction of parameters; full fine-tuning updates all weights.

MC4. (b) Precision = fidelity (generated samples on the real manifold); recall = coverage/diversity (real distribution covered by the generated samples).

MC5. (c) L-systems (Lindenmayer systems): parallel rule rewriting that models biological growth.

MC6. (b) DreamBooth fine-tunes the model weights (heavier, higher fidelity); Textual Inversion freezes the model and learns a single new embedding (lightweight, $\sim$ KB).

MC7. (b) A dedicated language encoder (T5) gives deeper language understanding (complex prompts, compositionality, typography) that CLIP’s image-aligned contrastive text encoder handles poorly.

MC8. (a) FID fits Gaussians to Inception features of real and generated images and compares their means and covariances (Fréchet distance). Note (c) describes KID, not FID.

MC9. (b) Lower temperature scales up the logits before softmax, sharpening the distribution toward high-probability tokens, more deterministic, less diverse (temperature $\rightarrow 0$ approaches greedy).

MC10. (b) GAN inversion finds the latent code that makes a *pretrained* generator reconstruct a given real image, mapping the image into latent space (typically as a first step before editing).

Model Answers (theoretical questions)

D1: Reconstruction vs. quality vs. diversity

The key split is **paired vs. distributional**.

- **Reconstruction:** per-instance, paired fidelity to a *specific* input (x vs. $\hat{x}$). Metric: **LPIPS** (or PSNR/SSIM/MSE).
- **Sample quality (fidelity):** realism of generated samples; do they lie on the data manifold? Distributional, unpaired. Metric: **FID** (or precision).
- **Diversity (coverage):** does the model cover the full variety of the data, or collapse to a few modes? Metric: **Recall** (Kynkäänniemi precision/recall).

FID conflates quality and diversity into one number; precision/recall separates them (precision = fidelity, recall = coverage).

D2: Diffusion stability and sampling speed

More stable than GANs because diffusion optimizes a single, stationary regression objective (predict the added noise ϵ) with ordinary gradient descent, whereas a GAN is a minimax game between two networks, a non-stationary moving target that causes oscillation, vanishing gradients, and mode collapse. No adversary $\Rightarrow$ stable.

Slower to sample because generation is iterative: many sequential denoising steps (~ 1000 , ~ 50 with DDIM), each a full forward pass, and the steps cannot be parallelized. GANs/VAEs generate in a single pass. (Mitigated by consistency models / LCM and rectified flow / distillation.)

D3: MMDiT vs. U-Net (FLUX, SD3)

- **Scalability:** transformers follow scaling laws far better than convolutional U-Nets, so quality improves predictably with size/data.
- **Text-image fusion:** separate weights per modality with *joint* attention let text and image attend bidirectionally, vs. the U-Net’s one-way cross-attention injection. This drives prompt adherence and typography.
- **Global attention:** self-attention captures long-range dependencies at every layer (better global coherence than local convolutions).

- **Uniformity:** a homogeneous transformer stack (no hand-designed multiresolution encoder/decoder), and it pairs naturally with rectified flow.

L1: Ways to control generative outputs

Control spans a spectrum: baked in at training $\rightarrow$ post-hoc on a frozen model $\rightarrow$ adapting the model to new concepts.

- **Control during training (cGAN).** Condition at training time: $G(z, c)$, $D(x, c)$, learning $p(x|c)$. Strong but inflexible: the conditioning is fixed and new conditions need retraining (e.g. Pix2Pix).
- **Post-hoc latent directions.** On a *frozen* GAN, find semantic directions and edit by walking along them ($w + \alpha n$). *Supervised:* InterFaceGAN (SVM boundary normal). *Unsupervised:* GANSpace (PCA), SeFa (closed-form eigendecomposition of layer weights).
- **GAN inversion and editing.** To edit a *real* image, invert it into the latent space (Image2StyleGAN by optimization; pSp/e4e encoders), then apply the directions above. Central issue: the *reconstruction-editability trade-off* (W vs. $W+$: $W+$ reconstructs better but lands off-manifold and edits worse).
- **Personalization.** Teach a frozen text-to-image diffusion model a new subject from a few images. *Textual Inversion:* freeze the model, learn a single new embedding ($\sim$ KB), lower fidelity. *DreamBooth:* fine-tune the weights with a class/prior-preservation loss, heavier, higher fidelity.

L2: Captioning vs. prompt inversion (and metrics)

Different objectives. Captioning produces a human-readable *description*; prompt inversion finds a prompt that, fed to a *specific* generator, *re-renders* an image close to the target. A prompt is judged by the image it produces, not by readability. Hence they are not equivalent: a good caption may not reproduce the image; the best inversion prompt may be gibberish (PEZ); inversion is model-specific; and there is no unique ground-truth prompt (many prompts $\rightarrow$ similar images).

Strategies (cheap $\rightarrow$ expensive): VLM captioning (BLIP, CLIP-Interrogator) for initialization; retrieval (CLIP nearest-neighbor in a corpus, re-ranked by rendered similarity); discrete prompt optimization (PEZ: optimize embeddings against CLIP, hard-project to tokens); soft-prompt inversion; LLM-based iterative refinement using image-side feedback; prompt sampling (render, score, keep).

Metrics for candidate prompts: image-side, since fluency alone is insufficient: **CLIP image-to-image similarity**, **LPIPS**, **pixel MSE**. For full marks, add a *non-CLIP* metric (**DINOv2** feature similarity) to avoid circularity with PEZ’s objective, and a **transferability** re-render with a different seed to detect fixed-seed overfitting (optionally prompt-readability via LM perplexity).